

Adavis AI Platform – Phase 1 Final Design Document

(Auth + License + MDM + RBAC | Multi-tenant | MongoDB)

1. ARCHITECTURE OVERVIEW

Microservices

API Gateway (Spring Cloud Gateway)

↓

Auth Service	→ Authentication, Tokens, Sessions
License Service	→ License validation, module enablement
MDM Service	→ Users, Groups, Roles, Permissions
Audit Service	→ Activity tracking (21 CFR compliance)

2. CORE DESIGN PRINCIPLES

- ✓ Multi-tenant (separate DB per tenant)
- ✓ MongoDB ObjectId as _id (DB internal only)
- ✓ Business identifiers (codes) for all references
- ✓ UI-driven access (Screen आधारित control)
- ✓ RBAC with hierarchy + deny override
- ✓ License-driven module visibility
- ✓ Full audit compliance

3. AUTHENTICATION DESIGN

Token Model

Token	Usage
auth_token	Used for all APIs

Token	Usage
refresh_token	Used only to regenerate auth_token

Token Lifecycle

Login
→ auth_token (15 min)
→ refresh_token (7-30 days)

API Calls → auth_token

Expiry → /auth/refresh → new tokens

4. LOGIN FLOW

Step 1: Initiate Login

POST /auth/login/initiate

```
{  
  "identifier": "admin@adavis.com"  
}
```

Response:

```
{  
  "userId": "ADMIN-001",  
  "email": "admin@adavis.com",  
  "status": "ACTIVE",  
  "passwordSet": true  
}
```

Step 2: Login

POST /auth/login

```
{
  "identifier": "admin@adavis.com",
  "password": "Admin@123"
}
```

Response:

```
{
  "authToken": "JWT",
  "refreshToken": "REFRESH",
  "userId": "ADMIN-001"
}
```

5. PASSWORD MANAGEMENT

 No Temporary Password

 Secure Token-Based Reset

Admin Trigger

```
POST /auth/password/admin-reset
```

Reset Token Collection

auth_password_tokens

```
{
  "_id": "ObjectId",
  "userId": "QA-001",
  "tokenHash": "HASH",
  "type": "PASSWORD_RESET",
}
```

```
"expiresAt": "ISODate",  
"isUsed": false,  
"createdAt": "ISODate"  
}
```

Set Password

```
POST /auth/password/set
```

```
{  
  "token": "RESET_TOKEN",  
  "newPassword": "Strong@123"  
}
```

6. AUTH COLLECTIONS

auth_users

```
{  
  "_id": "ObjectId",  
  "userId": "ADMIN-001",  
  "email": "admin@adavis.com",  
  "status": "ACTIVE",  
  "isLocked": false,  
  "failedAttempts": 0,  
  "lastLoginAt": "ISODate",  
  "isDeleted": false  
}
```

auth_credentials

```
{  
  "_id": "ObjectId",  
  "userId": "ADMIN-001",  
  "email": "admin@adavis.com",  
}
```

```
"passwordHash": "HASH",
"passwordLastChangedAt": "ISODate",
"forceReset": false
}
```

auth_sessions

```
{
  "_id": "ObjectId",
  "userId": "ADMIN-001",
  "refreshToken": "TOKEN",
  "ipAddress": "IP",
  "expiresAt": "ISODate",
  "isActive": true
}
```

7. USER PROFILE

mdm_user_profiles

```
{
  "_id": "ObjectId",
  "userId": "USR_001",
  "email": "user@company.com",
  "firstName": "John",
  "lastName": "Doe",
  "department": "Production",
  "designation": "Operator",
  "phoneNumber": "+123",
  "isActive": true,
  "isDeleted": false,
  "createdAt": "ISODate"
}
```

8. GROUP MANAGEMENT

mdm_groups

```
{
  "_id": "ObjectId",
  "groupCode": "GRP_PROD_LINEA",
  "groupName": "Line A Operators",
  "isActive": true
}
```

mdm_user_group_assignments

```
{
  "_id": "ObjectId",
  "userId": "USR_001",
  "groupCode": "GRP_PROD_LINEA",
  "isActive": true
}
```

mdm_group_role_assignments

```
{
  "_id": "ObjectId",
  "groupCode": "GRP_PROD_LINEA",
  "roleCode": "ROLE_OPERATOR",
  "isActive": true
}
```

9. 🤖 ROLE MANAGEMENT

mdm_roles

```
{
  "_id": "ObjectId",
  "roleCode": "ROLE_OPERATOR",
  "roleName": "Operator",
}
```

```
"parentRoleCode": "ROLE_SUPERVISOR",  
"hierarchyPath": ["ROLE_SUPERVISOR", "ROLE_OPERATOR"],  
"level": 3,  
"isActive": true  
}
```

10. MODULES / SCREENS / FEATURES

mdm_modules

```
{  
  "_id": "ObjectId",  
  "moduleCode": "MOD_MDM",  
  "moduleName": "Master Data Management",  
  "isActive": true,  
  "displayOrder": 1  
}
```

mdm_screens

```
{  
  "_id": "ObjectId",  
  "screenCode": "SCR_USER_MANAGEMENT",  
  "moduleCode": "MOD_MDM",  
  "screenName": "User Management",  
  "isActive": true,  
  "displayOrder": 1  
}
```

mdm_features

```
{  
  "_id": "ObjectId",  
  "featureCode": "FEAT_CREATE_USER",  
  "moduleCode": "MOD_MDM",  
  "featureName": "Create User",  
}
```

```
"isActive": true
}
```

11. ROLE PERMISSIONS

mdm_role_permissions

```
{
  "_id": "ObjectId",
  "roleCode": "ROLE_OPERATOR",
  "moduleCode": "MOD_MDM",
  "version": 1,
  "isActive": true,

  "screenPermissions": [
    {
      "screenCode": "SCR_USER_MANAGEMENT",
      "actions": ["READ"],
      "deniedActions": []
    }
  ],

  "featurePermissions": [
    {
      "featureCode": "FEAT_CREATE_USER",
      "actions": ["EXECUTE"]
    }
  ]
}
```

12. EFFECTIVE PERMISSIONS

mdm_user_effective_permissions

```
{
  "userId": "USR_001",
  "permissions": {
```



```
    "SCR_USER_MANAGEMENT": ["READ"]
  },
  "computedAt": "ISODate"
}
```

13. LICENSE MODULE

mdm_licenses

```
{
  "_id": "ObjectId",
  "plan": {
    "planId": "PLAN_ENTERPRISE",
    "planName": "Enterprise",
    "planType": "PAID"
  },
  "modules": ["MOD_MDM", "MOD_IOT"],
  "maxUsers": 200,
  "currentUsers": 120,
  "status": "ACTIVE",
  "startDate": "ISODate",
  "expiryDate": "ISODate"
}
```

14. AUDIT LOGS

mdm_audit_logs

```
{
  "_id": "ObjectId",
  "userId": "USR_001",
  "action": "CREATE_USER",
}
```

```
"entity": "USER",
"entityId": "USR_002",
"before": {},
"after": {},
"timestamp": "ISODate"
}
```

15. ADMIN ACTIONS

Activate User	→ PATCH /mdm/users/{id}/activate
Deactivate User	→ PATCH /mdm/users/{id}/deactivate
Block User	→ PATCH /auth/users/{id}/block
Unblock User	→ PATCH /auth/users/{id}/unblock
Delete User	→ DELETE /mdm/users/{id}
Admin Reset	→ POST /auth/password/admin-reset

16. COMPLETE ENDPOINT LIST

AUTH

```
POST /auth/login/initiate
POST /auth/login
POST /auth/refresh
POST /auth/logout
POST /auth/password/set
POST /auth/password/forgot
POST /auth/password/reset
POST /auth/password/admin-reset
GET /auth/me
```

USERS

```
POST /mdm/users
GET /mdm/users
GET /mdm/users/{id}
```

```
PUT /mdm/users/{id}
PATCH /mdm/users/{id}/activate
PATCH /mdm/users/{id}/deactivate
DELETE /mdm/users/{id}
```

GROUPS

```
POST /mdm/groups
GET /mdm/groups
POST /mdm/groups/{id}/users
DELETE /mdm/groups/{id}/users/{userId}
POST /mdm/groups/{id}/roles
DELETE /mdm/groups/{id}/roles/{roleId}
```

ROLES

```
POST /mdm/roles
GET /mdm/roles
PUT /mdm/roles/{id}
POST /mdm/roles/{id}/permissions
```

LICENSE

```
POST /license/activate
GET /license
GET /license/validate
```

AUDIT

```
GET /audit/logs
```

17. 🔥 FINAL PERMISSION FLOW

User

- Groups
- Roles
- Role Hierarchy
- Role Permissions
- Merge (ALLOW - DENY)
- License Filter
- FINAL ACCESS

18. 🚀 FINAL SUMMARY

- ✓ ObjectId-based Mongo schema
- ✓ Business-code driven relationships
- ✓ Enterprise RBAC (hierarchical)
- ✓ License-aware module control
- ✓ Secure authentication (JWT)
- ✓ Token-based password reset
- ✓ Multi-group & multi-role support
- ✓ Full audit compliance
- ✓ Bulk upload ready
- ✓ Production-ready